US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250275667
Kind Code	A1
Publication Date	September 04, 2025
Inventor(s)	PARK; Dong Hwi et al.

DISHWASHER

Abstract

A dishwasher includes a tub that defines a wash space configured to accommodate one or more objects to be washed, a sump that is configured to store wash water, a water supply pump that supplies the wash water to the tub, a base that is disposed below the tub, and provides a space to accommodate the sump and the water supply pump, an air supply device that is configured to supply air to the tub, a first filter that is disposed at the base and in fluid communication with the tub and the air supply device, a second filter that is disposed at the base and in fluid communication with the tub, and a water supply that is connected to the second filter and configured to supply the wash water.

Inventors: **PARK; Dong Hwi (Seoul, KR), KIM; Kyung Rae (Seoul, KR), KO; Myungwon (Seoul, KR)**

Applicant: **LG Electronics Inc.** (Seoul, KR)

Family ID: **82595219**

Appl. No.: **19/212863**

Filed: **May 20, 2025**

Foreign Application Priority Data

KR	10-2021-0091811	Jul. 13, 2021

Related U.S. Application Data

parent US continuation 17863725 20220713 parent-grant-document US 12342975 child US 19212863

Publication Classification

Int. Cl.: **A47L15/48** (20060101); **A47L15/42** (20060101); **A47L15/50** (20060101); **B01D46/00** (20220101); **B01D46/24** (20060101); **F26B21/00** (20060101)

U.S. Cl.:

CPC **A47L15/488** (20130101); **A47L15/4202** (20130101); **A47L15/4206** (20130101); **A47L15/4225** (20130101); **A47L15/4246** (20130101); **A47L15/4251** (20130101); **A47L15/486** (20130101); **A47L15/501** (20130101); **B01D46/0004** (20130101); **B01D46/0043** (20130101); **B01D46/2403** (20130101); **F26B21/003** (20130101); **F26B21/004** (20130101); B01D2273/30 (20130101); B01D2279/55 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
[0001] This application is a continuation of U.S. application Ser. No. 17/863,725, filed on Jul. 13, 2022, which claims priority to and the benefit of Korean Patent Application No. 10-2021-0091811, filed on Jul. 13, 2021, which are hereby incorporated by reference as when fully set forth herein.
TECHNICAL FIELD
[0002] The present disclosure relates to a dishwasher, and in particular, a dishwasher that provides the functions of hot air drying and filtering.
BACKGROUND
[0003] Dishwashers are devices that may spray water to objects to be washed such as cooking vessels, cooking tools, and the like stored therein and wash them. In some cases, detergent can be included in wash water used for a wash.
[0004] For instance, dishwashers may include a wash tub defining a wash space configured to receive objects to be washed in the wash tub, a spray arm for spraying wash water into the wash space, and a sump configured to store the wash water and supply the wash water to the spray arm.
[0005] Dishwashers may help users to spend less time and efforts in washing the dishes after meals, thereby improving user convenience.
[0006] In some cases, a dishwasher may include a discharge duct that is provided at a door and discharges air out of a tub.
SUMMARY
[0007] The present application describes a dishwasher that has a structure that can spray air, e.g., hot air, to objects to quickly dry the objects after a washing process is completed with wash water.
[0008] For example, the dishwasher can include an air spray structure configured to spray heated air smoothly into a tub that accommodates the objects, where the air spray can save space in the tub. To dry the objects quickly with air, the air spray structure can adjust a direction of the spray of air into the tub. To ensure sanitation, the air spray structure can filter foreign substances included in air spraying into the tub. Additionally, the air spray structure can enable a user to remove the collected foreign substances readily.
[0009] According to one aspect of the subject matter described in this application, a dishwasher comprises a tub that defines a wash space configured to accommodate one or more objects to be washed, a sump that is configured to store wash water, a water supply pump that supplies the wash water to the tub, a base that is disposed below the tub, and provides a space to accommodate the sump and the water supply pump, an air supply device that is configured to supply air to the tub, a first filter that is disposed at the base and in fluid communication with the tub and the air supply device, a second filter that is disposed at the base and in fluid communication with the tub, and a water supply that is connected to the second filter and configured to supply the wash water. The

first filter is disposed to be spaced from the sump on one lateral direction of the tub, and the second filter is disposed to be spaced from the sump on the other lateral direction of the tub.

[0010] The first filter comprises a casing that defines an accommodation space, an air filter disposed in the accommodation space of the casing and configured to filter foreign substances in air, and a cap mounted on the casing to replace the air filter, wherein the cap is disposed to protrude from a bottom surface of the tub to be exposed inside the tub.

[0011] The casing comprises a stopper that protrudes from a surface of the casing and has a ring shape and protrudes radially outward. The casing has a first side that is in fluid communication with the tub and a second side that is in fluid communication with the air supply device, and wherein the cap is detachably disposed at the casing. At least a portion of the casing is configured to be exposed to the bottom surface of the tub based on the cap being detached from the casing. The casing passes through a first penetration hole provided on the bottom surface of the tub, and at least a part of the casing is exposed.

[0012] The casing comprises an air suction part formed so that external air is suctioned in the radial direction of the air filter toward the center of the air filter, and wherein air suctioned in through the air suction part descends to a lower part of the air filter and is supplied to the air supply device.

[0013] The air supply device comprises a housing that defines a flow path configured to guide the air, the housing having an outlet configured to discharge the air from the flow path, an air blowing fan disposed in the housing and configured to cause the air to flow along the flow path, and a heater that is at least partially disposed in the housing and configured to heat the air blown by the air blowing fan. The first filter and the heater are in fluid communication with each other, and the air blowing fan is disposed between the first filter and the heater.

[0014] The tub further defines a first penetration hole at the bottom surface of the tub, the first filter being disposed in the first penetration hole, and a second penetration hole at the bottom surface of the tub, the second filter being disposed in the second penetration hole.

[0015] According to another aspect of the subject matter described in this application, a dishwasher comprises a tub that defines a wash space configured to accommodate one or more objects to be washed, a sump that is configured to store wash water, a water supply pump that supplies the wash water to the tub, a base that is disposed below the tub, and provides a space to accommodate the sump and the water supply pump, an air supply device that is disposed at the base and configured to supply air to the tub, and a guide vane that is configured to control a direction of the spray of air spraying to the tub. The tub comprises an opening into which air flows from the air supply device, and a first forming part which is depressed in the portion where the opening is formed, wherein the guide vane is disposed in the first forming part, and inserted into the opening.

[0016] The opening is provided in plurality, and each of the openings is arranged to be spaced from each other in the front-rear direction of the tub. The plurality of openings are disposed at the central portion of the side of the tub.

[0017] The dishwasher further comprises a lower rack that is disposed in the wash space and store object to be washed, an upper rack that is disposed over the lower rack and store object to be washed, a first holder that is disposed in the tub, and on which the lower rack is held, and a second holder that is disposed in the tub, disposed over the first holder, and on which the upper rack is held, wherein the opening is disposed between the first holder and the second holder. The tub further comprises a second forming part that is formed in the upper portion of the first forming part in a position where at least a portion of the second forming part overlaps the upper rack in a way that a inner surface of a lateral plate of the tub is depressed outward.

[0018] The air supply device comprises a housing that defines a flow path configured to guide the air, an air blowing fan disposed in the housing and configured to cause the air to flow along the flow path, and a heater that is at least partially disposed in the housing and configured to heat the air blown by the air blowing fan, wherein the dishwasher further comprises a guide duct that one side communicates with an outlet of the housing and the other side communicates with the

opening. The guide duct further comprises a division vane that is configured to divide at least a portion of a flow path that faces a plurality of openings, divides a position where each of the plurality of openings is disposed, and guide air to each of the plurality of openings.
[0019] The guide vanes and the openings are provided in plurality, and each of the plurality of guide vanes is mounted on each of the plurality of openings to change a direction of the flow of air flowing into the openings. The plurality of guide vanes change a direction of the flow of air flowing into the opening such that air sprays into the tub in the same direction. The plurality of guide vanes change a direction of the flow of air flowing into the opening such that air sprays into the tub in different directions.
[0020] In some implementations, Since a plurality of openings are provided, air may smoothly flow into the tub from the air supply device. In addition, since the plurality of openings are provided, the spray directions of an air flowing into the tub through each opening can be adjusted to be the same direction or different directions from each other.
[0021] In some implementations, where a first forming part is formed and accommodates a portion of a guide vane, protruding from the tub, a reduction in the storage space of the tub or the interference between the upper rack and the lower rack, which may be caused by the protruding portion of the guide vane, can be suppressed effectively.
[0022] In some implementations, the user can adjust a slant direction of wings of the guide vane to vary a direction of air flowing into the tub, considering wash conditions such as the amount, sorts and arrangement positions of objects to be washed placed in the tub, thereby enhancing efficiency of air drying.
[0023] In some implementations, the first filter can be coupled to the air supply device and have an air filter mounted thereon such that the air filter filters foreign substances such as dust and the like included in air, improving a sanitation level of the dishwasher.
[0024] In some implementations, where a distance between the first filter and the opening formed on a lateral plate of the tub is minimized, a flow path of air from the first filter to the opening can be simplified, thereby improving volumetric efficiency of the dishwasher and ensuring a simple structure of the dishwasher.
[0025] Specific effects are described along with the above-described effects in the section of detailed description.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0026] FIG. **1** is a schematic cross-sectional view showing an example of a dishwasher.
[0027] FIG. **2** is a perspective view showing an example of a coupling state of a tub and a base of the dishwasher.
[0028] FIG. **3** is a front cross-sectional view showing the dishwasher.
[0029] FIG. **4** is a perspective view showing a portion of the base in the dishwasher.
[0030] FIG. **5** is a perspective view of FIG. **4** from a different position.
[0031] FIG. **6** is a perspective view showing an example of a hot air spray part and a filter part that are coupled.
[0032] FIG. **7** is a cross-section view of FIG. **2**.
[0033] FIG. **8** is a view showing FIG. **7** without a guide vane.
[0034] FIG. **9** is a perspective view showing the hot air spray part in the dishwasher.
[0035] FIG. **10** is a perspective view showing an example of a hot air spray part.
[0036] FIG. **11** is an enlarged cross-sectional view showing a portion of the dishwasher.
[0037] FIG. **12** is a perspective view of FIG. **6** from a different position.
[0038] FIG. **13** is a view of FIG. **12** without a filter part.

[0039] FIG. **14** is a front view of FIG. **12**.
[0040] FIG. **15** is a view showing portion AA of FIG. **14**.
[0041] FIG. **16** is a view showing portion BB of FIG. **14**.
[0042] FIG. **17** is an exploded view of FIG. **12**.
[0043] FIG. **18** is a view of FIG. **17** from a different position.
[0044] FIG. **19** is a perspective view showing an example of an inside of the tub.
[0045] FIG. **20** is a perspective view showing an example of an inside of a tub.
[0046] FIG. **21** is a view showing example components of the filter part in FIG. **20** that are dissembled.
[0047] FIG. **22** is a perspective view showing an example of an air filter.
[0048] FIG. **23** is a perspective view showing an example of an air filter.

DETAILED DESCRIPTION

[0049] The above-described aspects, features and advantages are specifically described hereafter with reference to the accompanying drawings.
[0050] Throughout the disclosure, an "up-down direction" denotes an up-down direction of a dishwasher in a state in which the dishwasher is installed for use. A "left-right direction" denotes a direction orthogonal to the up-down direction, and a "front-rear direction" denotes a direction orthogonal to the up-down direction and the left-right direction. "Both lateral directions" or a "lateral direction" can have the same meaning as the left-right direction. These terms can be interchangeably used in the disclosure.
[0051] In the drawings, the z-axis direction can denote the up-down direction, the y-axis direction can denote both the lateral directions, the lateral direction or the left-right direction, and the x-axis direction can denote the front-rear direction.
[0052] FIG. **1** is a schematic cross-sectional view showing an example of a dishwasher **1**.
[0053] In some implementations, referring to FIG. **1**, the dishwasher **1** can include a housing forming the exterior of the dishwasher **1**, a tub **2** define a wash space **2***a* in the housing and accommodating an object to be washed, a door **3** selectively opening and closing the wash space **2***a*, a sump **4** being disposed in the lower portion of the tub **2** and storing wash water, a storage part **5** being disposed in the tub **2** and storing an object to be washed, and spray arms **6**, **7**, **9** spraying wash water to an object to be washed being stored in the storage part **5**. The object to be washed can be cooking vessels such as a bowl, a dish, a spoon, chop sticks and the like, and other cooking tools, for example. Hereafter, the object to be washed can be referred to as a cooking vessel.
[0054] The tub **2** can define the wash space **2***a* and accommodate cooking vessels, and the storage part **5** and the spray arms **6**, **7**, **9** can be provided in the wash space **2***a*. The tub **2** has a shape in which one surface is open, and the open one surface can be opened and closed by the door **3**.
[0055] The door **3** can be rotatably connected to the housing and configured to selectively open and close the wash space **2***a*. For example, the lower portion of the door **3** can be hinge-coupled to the housing.
[0056] In some implementations, the door **3** can rotate around a hinge, and open and close the tub **2**. When the door **3** is open, the storage part **5** can be withdrawn out of the dishwasher **1**, and the storage part **5** withdrawn outward can be supported by the door **3**.
[0057] The sump **4** can include a storage part **41** storing wash water, a sump cover **42** distinguishing the storage part **41** from the tub **2**, a water supply **43** supplying wash water to the storage part **41** from the outside, a water discharge part **44** discharging wash water of the storage part **41** outward, and a water supply pump **45** and a supply channel **46** for supplying wash water of the storage part **41** to the spray arms **6**, **7**, **9**.
[0058] The sump cover **42** can be disposed in the upper portion of the sump **4**, and distinguish the tub **2** from the sump **4**. Additionally, the sump cover **42** can be provided with a plurality of return holes for returning wash water having sprayed to the wash space **2***a* through the spray arms **6**, **7**, **9**.
[0059] That is, wash water having sprayed from the sprays arms **6**, **7**, and **9** can fall to the lower

portion of the wash space **2***a*, pass through the sump cover **42**, and return to the storage part **41** of the sump **4**.

[0060] The water supply pump **45** can be disposed in a lateral portion or the lower portion of the storage part **41**, and supply wash water to the spray arms **6**, **7**, **9** and the tub **2**.

[0061] One end of the water supply pump **45** can connect to the storage part **41**, and the other end can connect to the supply channel **46**. The water supply pump **45** can have an impeller **451**, a motor **453** and the like, therein. As the motor **453** is supplied with power, the impeller **451** can rotate, and wash water in the storage part **41** can be supplied to the spray arm **6**, **7**, **9** through the supply channel **46**.

[0062] The supply channel **46** can selectively supply wash water supplied from the water supply pump **45** to the spray arm **6**, **7**, **9**.

[0063] The supply channel **46** can include a first supply channel **461** connecting to a lower spray arm **6**, a second supply channel **463** connecting to an upper spray arm **7** and a top nozzle or spray arm **9**, and a supply channel diverting valve **465** selectively opening and closing the supply channels **461**, **467**. The supply channel diverting valve **465** can control the supply channels **461**, **463** such that each of the supply channels **461**, **463** is opened consecutively or simultaneously.

[0064] At least one of storage parts **5** for storing cooking vessels can be provided in the wash space **2***a*. The dishwasher **1** is provided with two storage parts **5** is illustrated in FIG. **1**, but not limited.

[0065] For example, the dishwasher **1** can include a single storage part or three or more of storage parts. The number of the spray arms can vary depending on the number of the storage parts.

[0066] The storage part **5** can include a lower rack **51** and an upper rack **53** that are used for storing cooking vessels. The lower rack **51** can be disposed in the wash space **2***a* and store cooking vessels. The upper rack **53** can be disposed over the lower rack **51** and store cooking vessels. A top rack can be disposed in a space between the upper side of the upper rack **53** and the top nozzle or spray arm **9**, and the top rack can store cooking vessels.

[0067] The lower rack **51** can be disposed over the sump **4**, and the upper rack **53** can be disposed further upward than the lower rack **51**. The lower rack **51** and the upper rack **53** and the top rack can be withdrawn outward through one open surface of the tub **2**.

[0068] For instance, a rail-type holder can be disposed on the inner circumferential surface of the tub **2**, and a wheel is provided under the racks **51**, **53**. A user can withdraw the storage part **5** outward, to sore cooking vessels or take out cooking vessels having cleaned.

[0069] The spray arms can be disposed in the tub **2** and spray wash water toward cooking vessels in the storage part **5**. The spray arms can include a lower spray arm **6**, an upper spray arm **7** and a top nozzle or spray arm **9**.

[0070] The lower spray arm **6** can be rotatably provided under the lower rack **51**, and spray wash water to cooking vessels. The upper spray arm **7** can be rotatably provided between the lower rack **51** and the upper rack **53**, and spray wash water to cooking vessels.

[0071] The lower spray arm **6** can be rotatably mounted on the sump cover **42** and spray wash water to the cooking vessels stored in the lower rack **51**. The upper spray arm **7** can be disposed over the lower spray arm **6** and spray wash water to the cooking vessels stored in the upper rack **53**. The top nozzle or spray arm **9** can be disposed in the upper portion of the wash space **2***a* and spray wash water to the lower rack **51** and the upper rack **53**.

[0072] As described above, the first supply channel **461** can supply wash water to the lower spray arm **6**, and the second supply channel **463** can supply wash water to the upper spray arm **7** and the top nozzle or spray arm **9**.

[0073] Referring to FIG. **1**, the dishwasher **1** can include a base **8**. The base **8** can be disposed under the tub **2**, and a machine room is formed in the base **8**, and the tub **2** can be mounted on the base **8**. The base **8** can provide space accommodating the sump **4** the water supply pump **45**, and accommodating a pump, an air supply device **100**, a first filter **200** and various types of devices provided for the dishwasher **1**.

[0074] The base **8** can have an outer wall, support the dishwasher **1** entirely, and form space in which various types of devices are accommodated with the outer wall.
[0075] FIG. **2** is a perspective view showing an example of a coupling state between a tub **2** and a base **8** in the dishwasher **1**. FIG. **3** is a front cross-sectional view showing the dishwasher **1**. FIG. **4** is a perspective view showing the portion of the base **8** in the dishwasher **1**.
[0076] In some implementations, the dishwasher **1** can further include a first filter **200** and a second filter **300**.
[0077] The first filter **200** can be mounted on the base **8** and communicate with the tub **2**. The first filter **200** can pass through the bottom surface of the tub **2** and communicate with the tub **2**, and communicate with the air supply device **100**. The first filter **200** can suction air and filter foreign substances in the air. A specific structure of the first filter **200** is described hereafter with reference to the drawings.
[0078] The second filter **300** can be mounted on the base **8**, communicate with the tub **2**, and connect to the water supply **43** supplying wash water. Wash water can passes through the second filter **300** through the water supply **43**, flow into the tub **2**, and be stored in the sump **4**.
[0079] The second filter **300** can be provided as a filtering device that removes foreign substance included in the wash water flowing into the tub **2**. Alternatively, the second filter **300** can also be provided as a water softening device that changes wash water from hard water to soft water.
[0080] The first filter **200** and the second filter **300** can be spaced from each other, on a bottom plate **21** of the tub **2**. The sump **4** that stores wash water can be disposed between the first filter **200** and the second filter **300**. That is, the first filter **200** can be disposed to be spaced from the sump **4** on one lateral direction of the tub **2**, and the second filter **300** can be disposed to be spaced from the sump **4** on the other lateral direction of the tub **2**.
[0081] Air and wash water flow through the first filter **200** and the second filter **300** respectively. When the first filter **200** and the second filter **300** are adjacent to each other, air and wash water may be mixed and flow into the first filter **200** and the second filter **300**.
[0082] To suppress the flow of a mixture of air and wash water into both the first filter **200** and the second filter **300**, the first filter **200** and the second filter **300** can be separated and spaced from each other.
[0083] The first filter **200** and the second filter **300** can be mounted on the base **8**, and the sump **4** takes up a large space of the base **8**. In some examples, the first filter **200** and the second filter **300** can be disposed to avoid the sump **4**.
[0084] Since the base **8** is mostly occupied by the sump **4** and accommodates a pump, a PCB module and various types of devices, there is not enough space in the base **8**. Accordingly, the first filter **200** and the second filter **300** can hardly be disposed in the same area of the base, except for an area of the base **8** which is taken up by the sump **4**.
[0085] In the structure, when the second filter **300** is disposed at one side of the sump **4**, the first filter **200** can be disposed at the other side of the sump **4**, for example. For the above-mentioned reason, the sump **4** can be disposed between the first filter **200** and the second filter **300**, for example.
[0086] The tub **2** can have a first penetration hole **211** on which the first filter **200** is mounted, and a second penetration hole **212** on which the second filter **300** is mounted, on the bottom surface thereof.
[0087] The first filter **200** and the second filter **300** can be in fluid communication with the tub **2**. For example, the first penetration hole **211** and the second penetration hole **212** can be formed on the bottom surface, i.e., the bottom plate **21**, of the tub **2**. The first filter **200** can be mounted on the first penetration hole **211**, and the second filter **300** can be mounted on the second penetration hole **212**.
[0088] Since the sump **4** is disposed between the first filter **200** and the second filter **300**, the sump **4** can be disposed between the first penetration hole **211** and the second penetration hole **212**.

[0089] FIG. **5** is a perspective view of FIG. **4** viewed from a different position. FIG. **6** is a perspective view showing an example of an air supply device **100** and a first filter **200** that are coupled. FIG. **7** is a cross-section view of FIG. **2**. FIG. **8** is a view showing FIG. **7** without a guide vane **142**.
[0090] The tub **2** and a holder provided in the tub **2** are described with reference to FIGS. **7** and **8**.
[0091] The tub **2** can be formed into a box comprised of thin plates, and a lateral surface where the door **3** is disposed and which is opened and closed by the door **3** is open, while the remaining lateral surface is closed. The tub **2** can be formed stereoscopically based on the press or sheet metal processing. The tub **2** can include an opening **22***a* into which air flows from the air supply device **100**, and a first forming part **22***b* which is depressed in the portion where the opening **22***a* is formed.
[0092] The tub **2** can be comprised of a bottom plate **21**, a lateral plate **22**, an upper plate and a rear plate **24**. The bottom plate **21** can form the bottom surface of the tub **2** and have an opening communicating with the sump **4**. The lateral plate **22** can be coupled to the bottom plate **21**.
[0093] A pair of lateral plates **22** can be provided in a way that the lateral plates **22** bend from the bottom plate **21** and form the lateral walls of the tub **2**. The lateral plates **22** can be provided with an opening **22***a* which communicates with a guide duct **140** and into which air flows from the guide duct **140**.
[0094] The upper plate can connect the upper end portions of the lateral plates **22** to each other and form a ceiling part of the tub **2**. The rear plate **24** can connect the rear end portions of the lateral plates **22** to each other and face the door **3**.
[0095] A rail-type holder for holding a rack can be provided on the lateral plates **22** of the tub **2**, and a pair of holders can be provided and disposed to face each other on the pair of lateral plates **22**. A holder can be comprised of a first holder **20***a*, a second holder **20***b* and a third holder **20***c*.
[0096] The first holder **20***a* can be disposed in the lower portion of the tub **2**, the second holder **20***b* can be disposed over the first holder **20**, and the third holder **20***c* can be disposed over the second holder **20***b*. The first holder **20***a*, the second holder **20***b* and the third holder **20***c* can be spaced a proper distance from one another in the up-down direction of the tub **2**.
[0097] The first holder **20***a* can be formed in a way that a portion of the lateral plate **22** of the tub **2** protrudes in the front-rear direction of the tub **2**, and the lower rack **51** can be held in the first holder **20***a*. The lower rack **51** can be guided by the first holder **20***a* and move in the front-rear direction of the tub **2**.
[0098] The second holder **20***b* can be formed in a way that a rail-shaped bar is coupled to the lateral plate **22** of the tub **2** so that the lengthwise direction of the rail-shaped bar can be parallel with the front-rear direction of the tub **2**, and the upper rack **53** can be held in the second holder **20***b*. The upper rack **53** can be guided by the second holder **20***b* and move in the front-rear direction of the tub **2**.
[0099] Likewise, the third holder **20***c* can be formed in a way that a bar is coupled to the lateral plate **22** of the tub **2** so that the lengthwise direction of the bar can be parallel with the front-rear direction of the tub **2**, and the top rack can be held in the third holder **20***c*. The top rack can be guided by the third holder **20***c* and move in the front-rear direction of the tub **2**.
[0100] Hereafter, the opening **22***a* is specifically described with reference to FIG. **8**. The opening **22***a* can be formed into an approximate circle or oval, and a plurality of openings **22***a* can be provided. Each of the openings **22***a* can be spaced from each other in the front-rear direction of the tub **2**. Since the plurality of openings **22***a* are spaced from each other, air flowing into the tub **2** through the opening **22***a* can be diffused into the tub **2** effectively.
[0101] The centers of the openings **22***a* can be disposed at the same height in the up-down direction of the tub **2**, the plurality of openings **22***a* are disposed at the central portion of the side of the tub **2**.
[0102] In FIG. **8**, the plurality of openings **22***a* can be disposed on the lateral plate **22** of the tub **2** so that straight lines L**1** and L**2** from the bottom surface of the tub **2** to the centers of the two

openings **22***a* can have the same length or a similar length in the up-down direction of the tub **2**.
[0103] Additionally, a straight line connecting the centers of the plurality of openings **22***a* can be parallel with the front-rear direction of the tub **2**. In FIG. **8**, the straight line SL connecting the centers of the two openings **22***a* can be parallel with the front-rear direction, i.e., the x-axis direction, of the tub **2**.
[0104] In the above structure, air flowing into the tub **2** from the plurality of openings **22***a* can be uniformly distributed and diffused to the front and rear of the tub **2**.
[0105] As illustrated in FIG. **8**, the openings **22***a* can be disposed between the first holder **20***a* and the second holder **20***b*. Since the openings **22***a* are disposed at a position where the openings **22***a* avoids the first holder **20***a* and the second holder **20***b*, air flowing through the openings **22***a* can be smoothly diffused into the tub **2** without be interrupted by the first holder **20***a* and the second holder **20***b*.
[0106] The openings **22***a* can be disposed closer to the second holder **20***b* than to the first holder **20***a*. Referring to FIG. **8**, a length from the lowermost point of the upper end surface of the first holder **20***a* to the center of the opening **22***a* can be greater than a length from the lower end surface of the second holder **20***b* to the center of the opening **22***a*.
[0107] In the structure, the openings **22***a* can be disposed close to the central portion of the tub **2** in the up-down direction of the lateral plate **22** of the tub **2** without overlapping the upper rack **53**. Accordingly, air can be diffused uniformly to the upper portion and the lower portion of the tub **2**.
[0108] In some implementations, the dishwasher **1** can further include an air supply device **100**. Hereafter, the air supply device **100** and the first filter **200** are specifically described with reference to the drawings.
[0109] FIG. **9** is a perspective view showing an example state of an air supply device **100** in the dishwasher **1**. FIG. **10** is a perspective view showing an example state of an air supply device **100**.
[0110] FIG. **11** is an enlarged cross-sectional view showing an example portion of the dishwasher **1**. FIG. **12** is a perspective view of FIG. **6** viewed from a different position. FIG. **13** is a view of FIG. **12** without a first filter **200**.
[0111] The air supply device **100** can be mounted on the base **8**, communicate with the tub **2**, and spray air to the tub **2**.
[0112] In some implementations, the first filter **200** and the air supply device **100** can be coupled to each other. Further, the dishwasher **1** may not be provided with the first filter **200**. As illustrated in FIG. **13**, the first filter **200** and the air supply device **100** can separate from each other.
[0113] In some examples, where the air supply device **100** can only be mounted on the base **8**, while the first filter **200** is not, the air supply device **100** can only be mounted on the dishwasher **1**, without the first filter **200**.
[0114] In the case of a dishwasher without the first filter **200**, it can be difficult to effectively suppress the flow of foreign substances included in air, flowing into the air supply device **100**, into the air supply device **100** while the air flowing into the air supply device **100** flows smoothly.
[0115] When the dishwasher **1** is provided only with the air supply device **100**, a mesh structure can be mounted on an inlet **120***a* of the air blowing fan **120** to prevent the flow of relatively large foreign substances or a component mounted on the base **8** into the air blowing fan **120**.
[0116] Unless stated otherwise, the structure of the dishwasher **1** provided with both the first filter **200** and the air supply device **100** is described hereafter.
[0117] The air supply device **100** can spray cold air that is the flow of unheated air or hot air that is the flow of heated air. The air supply device **100** can spray cold air or hot air into the tub **2**, by controlling the operation of a heater **130** disposed at the air supply device **100**.
[0118] In the state in which a heating device does not operate, the air supply device **100** can spray cold air to the tub **2**, and in the state in which a heating device operates, the air supply device **100** can spray hot air to the tub **2**. Hereafter, the air supply device **100** spraying hot air is described.
[0119] The air supply device **100** can include a housing **110**, an air blowing fan **120**, a heater **130**.

A flow path through which air flows can be formed in the housing **110**. The air blowing fan **120** can be mounted on the housing **110** and force air flowing into the air blowing fan **120** to flow.

[0120] The air blowing fan **120** can be controlled and rotated by the controller provided at the dishwasher **1**. The air blowing fan **120** can be mounted on the housing **110**, and an air blowing fan bracket can be formed at the housing **110**. The air blowing fan bracket can have a corresponding structure to the shape of the air blowing fan **120** on which the air blowing fan **120** is rotatably mounted.

[0121] An inlet **120*a*** of the air blowing fan **120** can be formed in the housing **110**, and air can flow into the inlet **120*a*** of the air blowing fan **120** in a parallel direction with the rotation axis of the air blowing fan **120**. The inlet **120*a*** of the air blowing fan **120** can be formed in a way that a hole is made at the air blowing fan bracket formed in the casing **220**. The inlet **120*a*** of the air blowing fan **120** can communicate with the outlet of the first filter **200**, i.e., an outlet **222*a*** of a second cell **222** described below.

[0122] At least a portion of the heater **130** can be disposed in the housing **110** and heat the air that is forced to flow by the air blowing fan **120**. The heater **130** can be built into the housing **110**, and the air, forced to flow by the air blowing fan **120**, can be heated by the heater **130** and become hot air while flowing in the housing **110**.

[0123] The heater **130**, for example, can be an electric resistance heating coil, i.e., a sheath heater structure in which a coil receives electricity from a power source and heated, but not limited. The end portion of the heater **130** can be exposed to the outside of the housing **110** to electrically connect to the power source.

[0124] In the structure in which the first filter **200** and the air supply device **100** are coupled, the first filter **200** and the heater **130** can communicate with each other, and the air blowing fan **120** can be disposed between the first filter **200** and the heater **130**.

[0125] The dishwasher can include a guide duct **140** in which air flows. One side of the guide duct **140** can communicate with the outlet of the housing **110**, and the other side can communicate with the opening **22*a*** of the tub **2** in a lateral portion of the tub **2**. The guide duct **140** can guide the movement of hot air being discharged from the heater **130** and guide the hot air into the tub **2**.

[0126] The exterior of the guide duct **140** can be entirely formed in a way that the dishwasher **1** has a narrow left-to-right width and a wide front-to-rear width, while the guide duct **140** has space, where air flows, therein. Since the guide duct **140** has the above-described exterior structure, the guide duct **140** can be mounted without difficulty on a portion between the outside of the tub **2** with little space in the left-right direction and the exterior material for the dishwasher **1**.

[0127] The tub **2** can be provided with a plurality of openings **22*a*** into which air flows from the air supply device **100**. The opening **22*a*** can be formed in a way that one side of the lateral plate **22** of the tub **2** penetrates. The opening **22*a*** can be provided on the lateral plate **22** and communicate with the guide duct **140**.

[0128] An air drying process can be performed in the state in which the door **3** is partially open. To reduce the risk caused by hot air discharged outward in the state in which the door **3** is open and imposed to the user or to prevent deterioration of the efficiency of air drying, the openings **22*a*** can be disposed on the lateral plate **22** of the tub **2** rather than the rear plate **24** of the tub **2** or the door **3**, for example.

[0129] When the openings **22*a*** are disposed at the door **3**, it is not easy to provide a flow path in which air is sent to the door **3** that moves. In this context, the openings **22*a*** can be disposed on the lateral plate **22** of the tub **2** rather than the door **3**, for example.

[0130] A plurality of openings **22*a*** into which air flows can be provided. In FIG. **7** and the like, two openings **22*a*** are provided, for example. In some examples, three or more openings **22*a*** can be provided.

[0131] Since the plurality of openings **22*a*** is provided, air can smoothly flow into the tub **2** from the air supply device **100**. Additionally, since the plurality of openings **22*a*** is provided, the direction in which air flowing into the tub **2** sprays can be adjusted to the same direction or a different direction through each opening in the portion of the opening **22***a*.

[0132] Further, the openings **22***a* can be disposed in a position where the spray of air is not interrupted such that air flows into the tub **2** smoothly, for example. Furthermore, hot air can spray closely to a large number of objects to be washed in the state in which the hot air is still hot and does not cool, for example.

[0133] For example, the openings **22***a* can be disposed to avoid the position where the openings **22***a* overlap the first holder **20***a*, the second holder **20***b* and the third holder **20***c*, for example. Additionally, the openings **22***a* can be disposed closer to the lower rack **51** and the upper rack **53** where a relatively large number of objects to be washed are placed than to the top rack, for example.

[0134] Since the openings **22***a* need to be disposed near the lower rack **51** and the upper rack **53**, the openings **22***a* can be disposed on the lateral plate **22** of the tub **2** than the upper plate of the tub **2**, for example.

[0135] Referring to FIG. **7**, the openings **22***a* can be disposed between the first holder **20***a* on which the lower rack **51** is mounted and the second holder **20***b* on which the upper rack **53** is mounted. That is, the openings **22***a* can be defined on the lateral plate **22** of the tub **2** at a position higher than the first holder **20***a* and lower than the second holder **20***b*.

[0136] The guide duct **140** can further include a vent hole **141** and a guide vane **142**. The vent hole **141** can communicate with the openings **22***a* such that hot air flowing from the heater **130** sprays to the tub **2**, and the number of the guide ducts **140** can be the same as that of the openings **22***a*.

[0137] The guide vane **142** can be mounted respectively on a plurality of vent holes **141** and control a direction of the spray of air spraying to the tub **2**. The guide vane **142** can be mounted in the vent hole **141**, and inserted into the opening **22***a* and provided in a way that at least a portion of the guide vane **142** is exposed to the lateral plate **22** of the tub **2**.

[0138] The guide vane **142** can have a predetermined thickness. Accordingly, at least a portion of the guide vane **142** can protrude from the inner surface of the lateral plate **22**. The protruding portion of the guide vane **142** can cause a reduction in the space of the tub **2**, where objects to be washed are stored, and interference with the upper rack **53** and the lower rack **51**.

[0139] To solve the problem, the tub **2** of the dishwasher **1** in some examples can be provided with a first forming part **22***b*. The first forming part **22***b* can be depressed and formed in the portion where the opening **22***a* is formed.

[0140] The first forming part **22***b* can be formed in the portion where the opening **22***a* is formed, in a way that the inner surface of the lateral plate **22** is depressed outward. Accordingly, the first forming part **22***b* on the inner surface of the tub **2** can have a depressed shape, and the guide vane **142** can be disposed in the depression.

[0141] Since the guide vane **142** is disposed in the first forming part **22***b*, the protruding portion of the guide vane **142** can be accommodated in the first forming part **22***b*. Thus, the guide vane **142** may not protrude on the inner surface of the lateral plate **22** except for the first forming part **22***b*.

[0142] In the structure, the lower rack **51** and the upper rack **53** moving near the inner surface of the lateral plate **22** may not be interrupted by the guide vane **142**, causing no reduction in the volume of the lower rack **51** and the upper rack **53** and the space where objects to be washed are stored to avoid interference.

[0143] In some implementations, since the first forming part **22***b* is formed and accommodates the portion of the guide vane **142**, protruding from the tub **2**, a reduction in the storage space of the tub **2** or the interference between the upper rack **53** and the lower rack **51**, which are caused by the protruding portion of the guide vane **142**, can be suppressed effectively.

[0144] The opening **22***a* can be formed in a way that the first forming part **22***b* penetrates. In this case, the first forming part **22***b* can have an enough surface area to include the opening **22***a* in the position where the opening **22***a* is formed.

[0145] At least a portion of the second holder **20***b* can be disposed in a position where the second holder **20***b* overlaps the first forming part **22***b* in the lateral direction of the tub **2** since the degree of freedom of the shape and surface area of the first forming part **22***b* can improve and no problem is caused by the overlapping between the second holder **20***b* and the first forming part **22***b*.
[0146] A second forming part **22***c* can be formed in the upper portion of the first forming part **22***b*, on the lateral plate **22** of the tub **2**. The second forming part **22***c* can be formed in the upper portion of the first forming part **22***b* in a position where at least a portion of the second forming part **22***c* overlaps the upper rack **53** in a way that the inner surface of the lateral plate **22** of the tub **2** is depressed outward.
[0147] In some examples, the second forming part **22***c* can be formed to have a shape corresponding to the shape of the lateral surface of the upper rack **53** or to have a surface area greater than the surface area of the lateral surface of the upper rack **53**. Since the second forming part **22***c* is formed, the volume of the upper rack **53** can increase, and more objects to be washed can be stored in the upper rack **53**.
[0148] Additionally, since the second forming part **22***c* is formed, space can be formed between the lateral plate **22** of the tub **2** and the upper rack **53**, and the space can help to reduce friction between the upper rack **53** and the lateral plate **22** when the upper rack **53** moves in the front-rear direction of the tub **2**.
[0149] A third forming part **22***d* can be formed in the position on which the third holder **20***s* is mounted, on the lateral plate **22**. Unlike the first forming part **22***b* and the second forming part **22***c*, the third forming part **22***d* can be formed in a way that the inner surface of the lateral plate **22** protrudes toward the inside of the tub **2**, to mount the third holder **20***c* reliably.
[0150] FIG. **14** is a front view of FIG. **12**. FIG. **15** is a view showing portion AA of FIG. **14**. FIG. **16** is a view showing portion BB of FIG. **14**. FIG. **17** is an exploded view of FIG. **12**. FIG. **18** is a view of FIG. **17** viewed from a different position.
[0151] As illustrated in FIG. **18**, the guide vane **142** can be formed into a circle entirely to correspond to the opening **22***a* and the vent hole **141** that are formed into a circle. A plurality of guide vanes **142** can be provided, and each of the plurality of guide vanes **142** can be spaced from each other.
[0152] The guide vane **142** can include an edge part **142***a* and a wing **142***b*. The wing **142***b* can be fixed by the edge part **142***a* forming a circular edge, and the lengthwise direction of the guide vane **142** can be the diameter direction of the edge part **142***a*.
[0153] A plurality of wings **142***b* can be provided and spaced from each other. For example, a space between the plurality of wings **142***b* can define a passage through which air flows. Each wing **142***b* can have a length that decreases from the center of the edge part **142***a* toward the edge of the edge part **142***a*, to correspond to the shape of the circular edge part **142***a*.
[0154] In some examples, the wing **142***b* and the edge part **142***a* can be integrally formed. The guide vane **142** can be made of a plastic material and manufactured in injection molding, for example, but not limited.
[0155] The guide vane **142** is specifically described with reference to FIG. **11**. The arrows in FIG. **11** indicate directions of the flow of air. In some examples, a plurality of guide vanes **142** and a plurality of openings **22***a* can be provided. For example, each of the plurality of guide vanes **142** can be mounted on each of the plurality of openings **22***a* to change a direction of the flow of air flowing into the openings **22***a*.
[0156] As illustrated in FIG. **11**, a plurality of guide vanes **142** can be provided, spaced from each other, and respectively mounted on a plurality of openings **22***a*. The widthwise direction of the wing **142***b* can be at a slant with respect to a direction of the flow of air. Each of the plurality of guide vanes **142** can be mounted on each of the plurality of openings **22***a* and the vent hole **141**. In the structure, the guide vane **142** can change a direction of the flow of air.
[0157] A direction of the flow of air flowing into the guide vane **142** can change to the width

direction of the wing **142***b* that is disposed at a slant in the widthwise direction. In the enlarged cross-sectional view of FIG. **11**, since air flowing in a direction facing the guide vane **142** can be guided by the wing **142***b* while passing through the wing **142***b*, a direction of the flow of the air changes by a predetermined angle with respect to the primary direction of the flow of the air.
[0158] Since the slant direction of the wing **142***b* of each guide vane **142** mounted on each of the plurality of openings **22***a* is adjusted, a direction of the flow of air flowing into the tub **2** can be adjusted. Accordingly, the direction of the flow of the air can vary in the tub **2**.
[0159] The plurality of guide vanes **142** can change a direction of the flow of air flowing into the opening **22***a* such that the air sprays into the tub **2** in different directions. Description in relation to this is provided with reference to FIG. **9**. The arrows in FIGS. **9** and **10** indicate directions of the flow of air.
[0160] Referring to FIG. **9**, the wings **142***b* of two guide vanes **142** can be disposed at a slant in the front-rear direction of the tub **2**. In some cases, the wings **142***b* of the two guide vanes **142** can be disposed at a slant respectively in the rearward and forward directions of the tub **2**.
[0161] In the structure, air flowing into the tub **2** can be guided by the wing **142***b* of the guide vane **142**. Accordingly, air flowing into one opening **22***a* can move from the lateral plate **22** of the tub **2** to the rear, its flow direction can be changed by the rear plate **24**, and the air can move to the right of the tub **2**.
[0162] Additionally, air flowing into the other opening **22***a* can move from the lateral plate **22** of the tub **2** to the front, its flow direction can be changed by the door **3**, and the air can move to the right of the tub **2**.
[0163] Since a direction of the flow of air can change in each opening **22***a* as described above, the direction of the flow of the air can vary, enhancing efficiency of air drying in the entire tub **2**.
[0164] The plurality of guide vanes **142** can change a direction of the flow of air flowing into the opening **22***a* such that air sprays into the tub **2** in the same direction. Description in relation to this is provided with reference to FIG. **10**.
[0165] Referring to FIG. **10**, the wings **142***b* of two guide vanes **142** can be disposed to incline toward the lower side of the tub **2** in the same direction. In the structure, air flowing into the tub **2** can be guided by the wings **142***b* of the guide vanes **142**, flow into the two openings **22***a*, and move from the lateral plate **22** of the tub **2** to the lower side, its flow direction can be changed by the bottom plate **21**, and the air can move to the right of the tub **2**.
[0166] When the direction of the flow of the air is the same in each opening **22***a* as described above, efficiency of air drying for objects to be washed being placed in a specific portion in the tub **2** can improve.
[0167] Additionally, when the direction of the flow of the air is the same in each opening **22***a*, air coming out of the two openings **22***a* can be mixed. Accordingly, vortex can be actively formed in the air, and the vortex can exert a strong impact on cooking vessels and help to improve the efficiency of air drying for the objects to be washed. In this case, since the vortex has no directionality, the objects to be washed can be affected by the air in different directions.
[0168] In some implementations, a direction of the flow of air spraying from each opening **22***a* can be adjusted in different directions from the above-described directions.
[0169] For example, the guide vane **142** can be detachably mounted on the opening **22***a* and the vent hole **141**. The user can detach the guide vane **142** from the opening **22***a* and the vent hole **141**. Accordingly, air the flow path of which is changed by the guide vane **142**, or air which is not guided by the guide vane **142** and has no directionality can flow into the tub **2**.
[0170] Additionally, the guide vane **142** can be mounted on the opening **22***a* and the vent hole **141** in a way that the guide vane **142** can rotate with respect to the lateral plate **22** of the tub **2**. Accordingly, the user can rotate the plurality of guide vanes **142** to adjust a slant direction of the wings **142***b* of each guide vane **142**, such that a direction of the flow of air can vary based on the number of cases.

[0171] In some implementations, the user can adjust a slant direction of the wings **142***b* of the guide vane **142** to vary a direction of air flowing into the tub **2**, considering wash conditions such as the amount, sorts and arrangement positions of objects to be washed placed in the tub **2**, thereby enhancing efficiency of air drying.

[0172] The guide duct **140** can be provided with a bent part **140***a* and an expanded part **140***b*. The bent part **140***a* and the expanded part **140***b* can be integrally formed in injection molding, for example.

[0173] The bent part **140***a* has an inlet connecting to the outlet of the housing **110**, and a bent portion. The bent part **140***a* can be entirely bent and have a stereoscopic shape. Since the bent part **140***a* has a stereoscopic shape, the bent part **140***a* can allow the housing **110** and the opening **22***a* to communicate with each other even if the housing **110** and the opening **22***a* are spaced from each other in a three-dimensional manner.

[0174] The expanded part **140***b* can have an inlet that connects to the bent part **140***a*, and a vent hole **141** that is an outlet. The expanded part **140***b* can have a structure in which the surface area of the expanded part **140***b* increases from the inlet toward the vent hole **141**. Accordingly, the expanded part **140***b* can have an enough surface area for a plurality of vent holes **141** having a relatively large surface area to be formed.

[0175] Referring to FIG. **16**, the guide duct **140** can further include a division vane **143** that divides a flow path formed in the guide duct **140**. The division vane **143** can be provided to divide at least a portion of the flow path that faces the plurality of openings **22***a* and is formed in the guide duct **140** that is formed in the guide duct **140** facing the plurality of openings **22***a*.

[0176] The division vane **143** can divide the position where each of the plurality of openings **22***a* is disposed, and guide air to each of the openings **22***a*. The division vane **143** can be formed into a partition wall in at least a portion of the expanded part **140***b*, and divide at least a portion of the flow path formed in the expanded part **140***b* into a first area **140***b***-1** and a second area **140***b***-2**.

[0177] Each opening **22***a* can be formed in the first area **140***b***-1** and the second area **140***b***-2**. Air flowing in the guide duct **140** can be guided by an edge wall of the guide duct **140** and the division vane **143** and flow, and a plurality of outlets can be formed in the portion of the plurality of openings **22***a*. Thus, air may not be mixed.

[0178] That is, the division vane **143** can guide air to each opening **22***a*, and allow air to smoothly flow in the guide duct **140**. When three or more openings **22***a* are formed, two or more division vane **143** can be formed. That is, the number of the division vanes **143** can be one less than the number of the openings **22***a*.

[0179] The first filter **200** can be detachably disposed at the air supply device **100**, and include an air filter **210**, a casing **220**, and a cap **230**.

[0180] The air filter **210** can be accommodated in the casing **220** and filter foreign substances included in air. The air filter **210** can be detachably mounted on the first filter **200**. The air filter **210** can filter foreign substance particles such as dust and the like included in air. The air filter **210** can be made of a porous material, and can be a high efficiency particulate air (HEPA) filter or an ultra low penetration air (ULPA) filter, for example, but not limited.

[0181] The casing **220** can provide accommodation space. The casing **220** can have one side communicating with the tub **2** and the other side communicating with the air supply device **100**. The air filter **210** can be mounted on the casing **220**. Referring to FIGS. **17** and **18**, the air supply device **100** and the first filter **200** can be coupled each other such that the inlet **120***a* of the air blowing fan **120** formed at the air supply device **100** communicates with the outlet **222***a* of the second cell **222** of the first filter **200**.

[0182] Thus, foreign substances can be removed from air by the air filter **210** while the air passes through the first filter **200**, and the air flows into the air supply device **100** and then flows into the tub **2**. As a consequence, air used in the process of air drying is without foreign substances and that a high level of sanitation is ensured during a process of washing cooking vessels.

[0183] The casing **220** can penetrate the bottom surface of the tub **2**. The casing **220** can be disposed to penetrate the bottom surface of the tub **2** through a first penetration hole formed on the bottom plate **21** of the tub **2**. Accordingly, the cap **230** can be mounted in the upper portion of the casing **220** and exposed to the lower portion of the tub **2**. The cap **230** can be disposed to protrude from the bottom surface of the tub **2** to be exposed inside the tub **2**. The cap **230** can be mounted on the casing **220** to replace the air filter **210** by opening and closing the casing **220** in which the air filter **210** is accommodated.
[0184] The cap **230** can be mounted on the casing **220**, and open and close the first penetration hole **211** formed on the bottom surface of the tub **2**. The cap **230** can be detachably disposed at the casing **220**, mounted on the casing **220**, and close a portion that the casing **220** communicate with the tub **2**. The cap **230** can be screw-coupled to the casing **220**, and the user can rotate the cap **230** to mount the cap **230** on the casing **220** or to detach the cap **230** from the casing **220**.
[0185] The casing **220** can have one side communicating with the tub **2** and the other side communicating with the air supply device **100**, and include a first cell **221** and a second cell **222**. For example, the first cell **221** and the second cell **222** can be integrally manufactured in injection molding and the like.
[0186] The first cell **221** can have a first communication hole **221***a* which communicates with the tub **2** and on which the cap **230** is mounted. The second cell **222** can have one side communicating with the first communication hole **221***a*, and the other side communicating with the air supply device **100**. As described above, the outlet **222***a* of the second cell **222** can communicate with the inlet **120***a* of the air blowing fan **120**.
[0187] The first cell **221** can be formed into a cylinder having a hollow hole, and include a first coupling part **2211** and a second coupling part. The first coupling part **2211** and the second coupling part can be integrally formed, for example. The first coupling part **2211** can have a screw thread that is formed in the upper portion of first coupling part **2211**, disposed in the first penetration hole **211** formed on the bottom surface of the tub **2**, and screw-coupled to the cap **230**.
[0188] The casing **220** can include a stopper **2211***a* that protrudes from the surface of the casing **220** and limits a range of the downward movement, i.e., the z-axis movement, of the cap **230**. The stopper **2211***a* can be disposed at the first coupling part **2211**.
[0189] The stopper **2211***a* may have a ring shape and protrude radially outward. The stopper **2211***a* may protrude radially outward from a boundary between the first coupling part **2211** and the air suction part **2212** described below.
[0190] The first coupling part **2211** can be provided with the stopper **2211***a* that protrudes outward from its lower end, i.e., the portion where the first coupling part **2211** is coupled to an air suction part **2212**, in the diameter direction, and is formed into a ring. The stopper **2211***a* can stop the lower end of the cap **230** from moving downward from the stopper **2211***a*.
[0191] The air suction part **2212** can be disposed under the first coupling part **2211**, and have a plurality of air suction holes **2212***a* communicating with the base **8**. A plurality of air suction holes **2212***a* can be arranged in the circumferential direction and the lengthwise direction of the first coupling part **2211**.
[0192] The air filter **210** can be formed into a cylinder having a hollow hole to be mounted in the hollow hole of the first cell **221**. The air filter **210** can be mounted on the inner circumferential surface of the first cell **221** to close the air suction hole **2212***a*.
[0193] Air can be suctioned from the outer circumference of the air suction part **2212** into the air suction part **2212**, pass through the air filter **210**, move in the diameter direction of the air filter **210** toward the center of the air filter **210**, and the move down to the lower portion of the air filter **210**. Thus, external air can be suctioned in the radial direction of the air filter **210** toward the center of the air filter **210** through the air suction part **2212**, and air suctioned in through the air suction part **2212** may descend to a lower part of the air filter **210** and may be supplied to the air supply device **100**.

[0194] The air having moved to the lower portion of the air filter **210** can flow into the air blowing fan **120** of the air supply device **100**, and finally, spray to the tub **2**. As the air blowing fan **120** operates, air can flow into the air suction hole **2212***a* from the space of the base **8**, and the air having flown into the air suction hole **2212***a* can be filtered by the air filter **210**, and foreign substances such as dust and the like can be filtered.
[0195] When the cap **230** is detached from the first coupling part **2211** of the casing **220**, at least a portion of the casing **220** can be exposed to the bottom surface of the tub **2**. As the cap **230** is detached, the air filter **210** can also be exposed to the bottom surface of the tub **2**. Accordingly, the user can approach the air filter **210** easily. The user can detach the air filter **210** exposed to the bottom surface of the tub **2** easily from the first filter **200** or mount the air filter **210** in the first filter **200**, after detaching the cap **230**.
[0196] When the dishwasher **1** continues to be used, the air filter **210** can be contaminated by foreign substances such as dust and the like. Accordingly, the air filter **210** can be washed or replaced with a new one on a regular basis.
[0197] In some implementations, the user can detach the air filter **210** from the first filter **200** and mount the air filter **210** in the first filter **200** as follows. The user can rotate the cap **230** and detach the cap **230** from the first filter **200**. As the cap **230** is detached, the air filter **210** can be exposed to the bottom surface of the tub **2** at least partially. Accordingly, the user can pick up the air filter **210** and detach the air filter **210** from the first filter **200**.
[0198] The user can mount the detached air filter **210** that is washed or a new air filter **210** in the first communication hole **221***a* formed in the first cell **221** of the first filter **200**. Then the user can fit the cap **230** into the first coupling part **2211** of the first cell **221**, rotate the cap **230**, and close the first communication hole **221***a* to mount the air filter **210** in the first filter **200** again.
[0199] Referring to FIG. **15**, the first cell **221** can have a filter guide **80** on which the air filter **210** is mounted, in a portion of the lower portion thereof, which corresponds to the first communication hole **221***a*. The filter guide **80** can be disposed in the lower portion of the first communication hole **221***a* and support the air filter **210** disposed in the upper portion of the first communication hole **221***a*.
[0200] The filter guide **80** can be formed into a rib that protrudes upward in the first cell **221**. When the air filter **210** is mounted in the first cell **221**, the filter guide **80** can guide the air filter **210** such that the air filter **210** sits in a position where the air filter **210** can filters air flowing into the first cell **221**, and suppress the movement of the air filter **210** in the first cell **221**, thereby supporting the air filter **210** reliably.
[0201] Additionally, the height of the upper surface of the filter guide **80** can be properly determined such that the user picks up the air filter **210** mounted in the first communication hole **221***a* to replace the air filter **210** easily.
[0202] In some implementations, since the first filter **200** which is coupled to the air supply device **100** and in which the air filter **210** is mounted is provided, the air filter **210** filters foreign substance such as dust and the like included in air and help to improve the sanitation level of the dishwasher **1**.
[0203] FIG. **19** is a perspective view showing an example of an inside of the tub **2**. FIG. **20** is a perspective view showing an example of ab inside of a tub **2**. As described above, the dishwasher **1** can be provided only with the air supply device **100** or both of the air supply device **100** and the first filter **200**.
[0204] FIG. **19** shows a tub **2** of the dishwasher **1**, which is provide only with the air supply device **100** but for the first filter **200**. In some cases, since the first filter **200** is not provided, a structure for detaching and mounting the air filter **210** may not be included. Accordingly, a structure in relation to the first filter **200** may not be provided on the bottom plate **21** of the tub **2**.
[0205] FIG. **20** shows a tub **2** of the dishwasher **1**, which is provided with both the air supply device **100** and the first filter **200**. In this case, a structure for enabling the user to detach and mount

the air filter **210** readily can be provided.
[0206] For example, the first penetration hole **211** can be formed on the bottom plate **21** of the tub **2**, and the first cell **221** of the first filter **200** and the cap **230** coupled to the first cell **221** can be disposed at the first penetration hole **211**.
[0207] The positions where the first cell **221** and the cap **230** are disposed can be taken into account. There is not enough space for mounting the first filter **200** and the air supply device **100** in portions adjacent to the lateral plates **22**, the rear plate **24** and the upper plate of the tub **2**. The base **8** is disposed below the bottom plate **21** of the tub **2**, and enough space for mounting the casing **220** of the first filter **200** and the housing **110** of the air supply device **100** can be ensured in the base **8**.
[0208] Additionally, the bottom plate **21** of the tub **2** can be easily approached by the user. The user can easily detach the lower rack **51** from the tub **2**, and as the lower rack **51** is detached, the user can easily approach the bottom plate **21**.
[0209] Further, the user can see the bottom plate **21** easily when looking down. Accordingly, when replacing the air filter **210**, the user can easily find the first cell **221** in which the air filter **210** is mounted, and the cap **230** coupled to the first cell **221**, and have a comfortable posture and look at the air filter **210** while replacing the air filter **210**.
[0210] For the above reasons, the cap **230** and the first cell **221** can be disposed in the portion of the bottom plate **21**, and the first penetration hole **211** can also be formed on the bottom plate **21**, for example.
[0211] FIG. **21** is a view showing example components of the first filter **200** in FIG. **20** that are dissembled. As described above, the first filter **200** can be disposed under the bottom plate **21** of the tub **2**, and the first penetration hole **211**, in which the first cell **221** and the cap **230** are disposed, is formed on the bottom plate **21** of the tub **2**, for example, for user convenience.
[0212] Hereafter, a position relationship between the opening **22***a* formed on the lateral plate **22** of the tub **2**, and the portion where the first filter **200** penetrates the bottom plate **21** and is disposed on the bottom plate **21**, i.e., the first penetration hole **211** formed on the bottom plate **21** of the tub **2**, is taken into account. The position of the first penetration hole **211** corresponds to the position of the first cell **221** of the first filter **200**. Accordingly the position relationship between the opening **22***a* and the first penetration hole **211** can be withdrawn based on the position relationship between the opening **22***a* and the first cell **221**.
[0213] Air used for air drying can flow into the first cell **221** of the first filter **200**, be heated while flowing in the air supply device **100**, pass through the guide duct **140** of the air supply device **100** and spray to the tub **2** through the opening **22***a*.
[0214] The length and shape of the bent part **140***a* of the guide duct **140** can be adjusted based on the arrangement position and a distance between the first cell **221** and the opening **22***a*, to allow the first cell **221** and the opening **22***a* to communicate with each other. As the distance between the first cell **221** and the opening **22***a* increases, the first cell **221** and the opening **22***a* can hardly communicate with each other.
[0215] The sump **4** storing wash water can be provided at the center of the bottom plate **21**. The sump **4** can occupy large space on the bottom plate **21**, and it is difficult to dispose pipes and components in a portion of the bottom plate **21**, where the sump **4** is disposed, below the bottom plate **21**.
[0216] If the opening **22***a* is far from the first cell **221** with respect to the center of the bottom plate **21**, the bent part **140***a* is disposed to avoid the sump **4** occupying the center of the bottom plate **21** across the bottom plate **21**.
[0217] In some cases, the bent part **140***a* forming a flow path of air may have a complex shape, making it difficult to design the dishwasher **1** as a whole. To simply design the shape of the bent part **140***a*, the opening **22***a* and the first cell **221** can be disposed close to each other with respect to the center of the bottom plate **21**, i.e., in the same area, for example.
[0218] Since the first cell **221** and the first penetration hole **211** are disposed in the same position,

the first penetration hole **211** and the opening **22***a* can be disposed in the same area with respect to the center of the bottom plate **21**, for example.

[0219] When the tub **2** is divided in a parallel direction with the direction where the lateral plate **22** is disposed, with respect to the center of the bottom plate **21** of the tub **2**, the opening **22***a*, and the portion where the first filter **200** penetrates the bottom plate **21** and is disposed on the bottom plate **21**, i.e., the first penetration hole **211**, can be disposed in the same area, for example.

[0220] FIG. **22** is a perspective view showing an air filter **210**. FIG. **23** is a perspective view showing an example of an air filter **210**.

[0221] In some implementations, referring to FIG. **22**, the air filter **210** can be formed into a cylinder having a hollow hole. The thickness of the air filter **210**, i.e., a distance between the outer circumferential surface and the inner circumferential surface of the air filter **210**, can be properly determined based on a material for the air filter **210**.

[0222] Referring to FIG. **23**, the air filter **210** can be formed into a cylinder having a hollow hole, and can be formed in a way that a concave and convex part **210***a* is formed in the circumferential direction on at least any one of the inner circumference and the outer circumference of the air filter **210**.

[0223] The inner circumference and the outer circumference of the filter can be surfaces through which air passes. Since the concave and convex part **210***a* is formed on the inner circumference and/or the outer circumference of the air filter **210**, the surface area of the air filter **210**, through which air passes, expands, thereby enhancing the air filter **210**'s effect of filtering foreign substances.

[0224] In some implementations, the first filter **200** can be disposed in a position easily approached by the user, and the user can easily detach and mount the air filter **210**, providing convenience of washing or replacing the air filter **210**.

[0225] Since a distance between the first filter **200**, and the opening **22***a* formed on the lateral plate **22** of the tub **2** is minimized, a flow path of air from the first filter **200** to the opening **22***a* can be simplified, thereby improving volumetric efficiency of the dishwasher **1** and ensuring a simple structure of the dishwasher **1**.

[0226] The implementations are described above with reference to a number of illustrative implementations thereof. However, implementations are not limited to the implementations and drawings set forth herein, and numerous other modifications and implementations can be devised by one skilled in the art within the technical scope of the disclosure. Further, the effects and predictable effects based on the configurations in the disclosure are to be included within the range of the disclosure though not explicitly described in the description of the implementations.

Claims

1. A dishwasher, comprising: a tub that defines a wash space configured to accommodate one or more objects to be washed; a sump that is configured to store wash water; a base that is disposed below the tub, and provides a space to accommodate the sump; an air supply device that is configured to supply air to the tub; a first filter that is disposed at the base and in fluid communication with the tub and the air supply device, the first filter being configured to filter the air; and a second filter that is disposed at the base and in fluid communication with the tub, the second filter being configured to filter the wash water, wherein the first filter comprises: a casing that defines an accommodation space, an air filter disposed in the accommodation space of the casing and configured to filter foreign substances in air, and a cap mounted on the casing to replace the air filter and disposed to be exposed inside the tub, and wherein the tub further defines: a first penetration hole at a bottom surface of the tub, the first filter being disposed in the first penetration hole; and a second penetration hole at the bottom surface of the tub, the second filter being disposed in the second penetration hole.

2. The dishwasher of claim 1, further comprising: a water supply pump configured to supply the wash water to the tub; and a water supply that is connected to the second filter and configured to supply the wash water.
3. The dishwasher of claim 1, wherein the casing comprises a stopper that protrudes from a surface of the casing and has a ring shape and protrudes radially outward.
4. The dishwasher of claim 1, wherein the casing has a first side that is in fluid communication with the tub and a second side that is in fluid communication with the air supply device, and wherein the cap is detachably disposed at the casing.
5. The dishwasher of claim 1, wherein at least a portion of the casing is configured to be exposed to the bottom surface of the tub based on the cap being detached from the casing.
6. The dishwasher of claim 1, wherein the casing passes through the first penetration hole provided on the bottom surface of the tub, and at least a portion of the casing is exposed.
7. The dishwasher of claim 1, wherein the casing comprises an air suction part formed so that external air is suctioned in a radial direction of the air filter toward the center of the air filter, and wherein air suctioned in through the air suction part descends to a lower part of the air filter and is supplied to the air supply device.
8. The dishwasher of claim 1, wherein the air supply device comprises: a housing that defines a flow path configured to guide the air, the housing having an outlet configured to discharge the air from the flow path; an air blowing fan disposed in the housing and configured to cause the air to flow along the flow path; and a heater that is at least partially disposed in the housing and configured to heat the air blown by the air blowing fan.
9. The dishwasher of claim 8, wherein the first filter and the heater are in fluid communication with each other, and the air blowing fan is disposed between the first filter and the heater.
10. A dishwasher, comprising: a tub that defines a wash space configured to accommodate one or more objects to be washed; a sump that is configured to store wash water; a base that is disposed below the tub, and provides a space to accommodate the sump; an air supply device that is configured to supply air to the tub; and a first filter that is disposed at the base and in fluid communication with the tub and the air supply device, the first filter being configured to filter the air, wherein the first filter comprises: a casing that defines an accommodation space, an air filter disposed in the accommodation space of the casing and configured to filter foreign substances in air, and a cap mounted on the casing to replace the air filter and disposed to be exposed inside the tub.
11. The dishwasher of claim 10, further comprising a second filter that is disposed at the base and in fluid communication with the tub, the second filter being configured to filter the wash water.
12. The dishwasher of claim 10, wherein the casing comprises a stopper that protrudes from a surface of the casing and has a ring shape and protrudes radially outward.
13. The dishwasher of claim 10, wherein the casing has a first side that is in fluid communication with the tub and a second side that is in fluid communication with the air supply device, and wherein the cap is detachably disposed at the casing.
14. The dishwasher of claim 11, wherein the tub further defines: a first penetration hole at a bottom surface of the tub, the first filter being disposed in the first penetration hole; and a second penetration hole at the bottom surface of the tub, the second filter being disposed in the second penetration hole.
15. The dishwasher of claim 10, wherein the casing comprises an air suction part formed so that external air is suctioned in a radial direction of the air filter toward the center of the air filter, and wherein air suctioned in through the air suction part descends to a lower part of the air filter and is supplied to the air supply device.
16. A dishwasher, comprising: a tub that defines a wash space configured to accommodate one or more objects to be washed; a base that is disposed below the tub; an air supply device that is configured to supply air to the tub; and a first filter that is disposed at the base and in fluid

communication with the tub and the air supply device, the first filter being configured to filter the air, wherein the first filter comprises: a casing that defines an accommodation space, an air filter disposed in the accommodation space of the casing and configured to filter foreign substances in air, and a cap mounted on the casing to replace the air filter and disposed to be exposed inside the tub, and wherein the casing passes through a first penetration hole provided on a bottom surface of the tub, and at least a portion of the casing is exposed.